

STIM1 Deficiency (CRAC Channelopathy)

— Comprehensive Disease Characterization Report

Disease: STIM1 Deficiency **MONDO ID:** MONDO:0013008 **OMIM:** #612783 (Immunodeficiency 10, IMD10) **Category:** Mendelian, autosomal recessive **Gene:** *STIM1* (Stromal Interaction Molecule 1; HGNC:11386; NCBI Gene 6786; OMIM *605921; chromosome 11p15.4)

Summary

STIM1 deficiency is an **ultra-rare, autosomal-recessive CRAC (Ca²⁺-release-activated Ca²⁺) channelopathy** caused by **biallelic loss-of-function (LOF) variants in *STIM1***, the endoplasmic-reticulum (ER) Ca²⁺ sensor that activates the plasma-membrane channel ORAI1. When STIM1 cannot sense ER Ca²⁺ depletion or engage ORAI1, **store-operated Ca²⁺ entry (SOCE)** is abolished. Because SOCE is a near-universal cellular signaling module, its loss produces a **congenital multisystem syndrome**: a SCID-like combined immunodeficiency accompanied — paradoxically — by autoimmunity and lymphoproliferation, together with non-immune features including muscular hypotonia/myopathy, anhidrotic (anhidrotic) ectodermal dysplasia with defective sweating, dental enamel hypomineralization, and pupillary/iris abnormalities (mydriasis). The syndrome is shared with recessive ORAI1 deficiency, its molecular partner, and together they define "CRAC channelopathy" ([PMID: 26469693](#)).

The **core causal chain** is well established: biallelic LOF *STIM1* mutation → loss/nonfunction of STIM1 protein → failure of the luminal EF-hand/SAM (EF-SAM) domain to sense ER Ca²⁺ depletion and oligomerize → failure to translocate to ER–plasma-membrane junctions and gate ORAI1 via the CRAC activation domain (CAD) → CRAC channels remain closed → absent SOCE → collapse of the downstream **Ca²⁺–calmodulin–calcineurin–NFAT** transcriptional axis in lymphocytes (impaired cytokine production despite normal lymphocyte development) plus failure of Ca²⁺-dependent functions in muscle, sweat gland, ameloblast, and iris smooth muscle. Critically, STIM1 deficiency (LOF) is the **mechanistic mirror image** of dominant

STIM1 gain-of-function (GOF), which causes constitutive SOCE and the tubular aggregate myopathy (TAM) / Stormorken syndrome spectrum — a distinction essential for correct classification and rational therapy.

The **only curative therapy for the immunodeficiency is allogeneic hematopoietic stem cell transplantation (HSCT)**, as for other combined immunodeficiencies; HSCT does not correct the non-hematopoietic (muscle, ectodermal, dental) features, which are managed supportively. A **splice-correcting antisense oligonucleotide (ASO)** that restores STIM1 splicing/function in patient cells has been demonstrated as a proof-of-concept, mutation-specific therapy ([PMID: 38977117](#)). Untreated, the disease is life-threatening in infancy/early childhood from recurrent, severe, and opportunistic infections, compounded by autoimmune cytopenias and lymphoproliferation.

1. Disease Information

STIM1 deficiency is a **primary (inborn) error of immunity** classified as a CRAC channelopathy. It is defined by the loss of store-operated Ca^{2+} entry (SOCE) secondary to biallelic loss-of-function of the ER Ca^{2+} sensor STIM1. As stated in the landmark review, "**CRAC channelopathy is caused by loss-of-function mutations in ORAI1 and STIM1 that abolish CRAC channel function and SOCE; it is characterized by severe combined immunodeficiency (SCID)-like disease, autoimmunity, muscular hypotonia, and ectodermal dysplasia, with defects in sweat gland function and dental enamel formation**" ([PMID: 26469693](#)).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0013008
OMIM	#612783 (Immunodeficiency 10)
Gene (HGNC)	STIM1, HGNC:11386
NCBI Gene	6786 (human); 20866 (mouse <i>Stim1</i>)
UniProt	Q13586 (human STIM1)
Orphanet	CRAC channelopathy spectrum (immunodeficiency by defective SOCE)
ICD-11	4A00 (immunodeficiencies) group
MeSH	Related terms: "Severe Combined Immunodeficiency"; "Stromal Interaction Molecule 1"

Synonyms / alternative names: Immunodeficiency 10 (IMD10); STIM1 loss-of-function; CRAC channelopathy (STIM1 type); combined immunodeficiency with autoimmunity due to STIM1 deficiency; store-operated calcium entry (SOCE) deficiency.

Source of information: The disease is characterized almost entirely from **individual patient reports and small consanguineous kindreds** (fewer than ~15 families reported worldwide) combined with **in vitro functional studies and animal models** — not from aggregated EHR-scale registries. The evidence base is therefore case-based human clinical data plus mechanistic model-organism and cellular studies.

2. Etiology

Primary cause — purely genetic. STIM1 deficiency is caused **solely by biallelic recessive loss-of-function variants in *STIM1*** that abolish SOCE. There are **no environmental, infectious, or toxic causes**, no somatic contribution, and no established modifier genes or disease-specific epigenetic changes. Infections in patients are downstream *consequences* of the immunodeficiency, not causes (Finding F012).

Genetic risk factors. The causal genetic events are germline biallelic LOF variants (homozygous or compound heterozygous). Reported variants include nonsense/frameshift alleles (e.g., c.685delT, p.Phe229Leufs12, *causing complete protein loss*; [PMID: 33733462](#)) and splice-site variants (e.g., NM_003156 c.792-3C>G *producing exon-7 skipping/intron retention with impaired SOCE*; [PMID: 38977117](#)). *Consanguinity is a strong risk factor**, as expected for a rare recessive disorder — e.g., "we studied two siblings from a consanguineous Syrian family" ([PMID: 33733462](#)).

Environmental / lifestyle risk factors: None identified. **Protective factors (genetic or environmental):** None established.

Gene–environment interactions: None mechanistically established. The only "interaction" is that pathogen exposure unmasks and drives the clinical immunodeficiency, but pathogens are not co-causal.

LOF vs GOF dichotomy (etiologic classification). STIM1 deficiency (recessive LOF) is the mechanistic opposite of autosomal-dominant STIM1 **gain-of-function**, which causes constitutive CRAC activation and the TAM/Stormorken spectrum: "**By contrast, autosomal dominant gain-of-function mutations in ORAI1 and STIM1 result in constitutive CRAC channel activation, SOCE, and increased intracellular Ca²⁺ levels that are associated with an overlapping spectrum of diseases, including nonsyndromic tubular aggregate myopathy (TAM) and York platelet and Stormorken syndromes**" ([PMID: 26469693](#)). "**Loss- and gain-of-function gene mutations in ORAI1 and STIM1 in human patients cause distinct disease syndromes**" ([PMID: 26469693](#)).

3. Phenotypes

STIM1 deficiency is a **congenital multisystem disorder**. The immunological phenotype (combined immunodeficiency + immune dysregulation) is essentially universal; individual non-immune features are variably present (variable expressivity). Frequencies are qualitative given the very small number of reported patients (F011).

Phenotype	Type	HPO term	Onset	Frequency
Combined immunodeficiency (SCID-like)	Clinical/ lab	HP:0005387	Congenital/ infantile	Near-universal
Recurrent/opportunistic infections	Clinical	HP:0002719	Infantile	Near-universal
Autoimmune hemolytic anemia	Lab/ clinical	HP:0001890	Infantile/ childhood	Common
Autoimmune thrombocytopenia	Lab/ clinical	HP:0001973	Infantile/ childhood	Common
Lymphoproliferation / lymphadenopathy	Clinical	HP:0002716	Childhood	Common
Hepatosplenomegaly	Clinical	HP:0001433	Childhood	Variable
Muscular hypotonia	Physical	HP:0001252	Congenital	Common
Muscle weakness / myopathy	Physical	HP:0001324	Congenital	Common
Hypohidrosis / anhidrosis	Physical	HP:0000970 / HP:0009925	Congenital	Common (ectodermal dysplasia)
Dental enamel hypoplasia / amelogenesis imperfecta	Physical	HP:0006297 / HP:0000705	Congenital (dentition)	Common
Mydriasis / pupillary abnormality	Physical	HP:0000535	Congenital	Reported
Skin hyperlaxity / elastic skin	Physical	—	Congenital	Reported (expanded phenotype)
Dysmorphic facies, hypoplastic patellae	Physical	—	Congenital	Reported (expanded phenotype)

Key supporting quotes: "in the case of STIM1 deficiency, autoimmunity and lymphoproliferative disease. The immunodeficiency in these patients is due to a severe defect in T cell activation but not in lymphocyte development" (PMID: 20189884); the

disease "is dominated by severe immunodeficiency and autoimmunity due to impaired SOCE" (PMID: 22615435); "muscular hypotonia, and ectodermal dysplasia, with defects in sweat gland function and dental enamel formation" (PMID: 26469693); the expanded phenotype "presenting with muscle weakness, hyperlaxity, elastic skin, tooth abnormalities, dysmorphic facies, hypoplastic patellae and history of respiratory infections" (PMID: 33733462).

Quality-of-life impact: Severe. Life-threatening infections dominate infancy; chronic autoimmune cytopenias require transfusion/immunosuppression; anhidrosis causes heat intolerance and hyperthermia risk; enamel defects affect dentition and nutrition; hypotonia/myopathy impairs motor development. Formal EQ-5D/SF-36 data are not available for this ultra-rare disease.

Severity/progression: Immune features are severe and life-threatening but treatable by HSCT; non-immune features are largely **congenital and static/non-progressive** rather than degenerative.

4. Genetic / Molecular Information

Causal gene: *STIM1* (HGNC:11386; NCBI Gene 6786; OMIM *605921; UniProt Q13586), encoding the single-pass ER-membrane Ca²⁺ sensor Stromal Interaction Molecule 1.

Pathogenic variants (biallelic, recessive):

Variant (cDNA / protein)	Type	Consequence	Reference
c.685delT, p.Phe229Leufs*12 (homozygous)	Frameshift	Complete loss of STIM1 protein	PMID: 33733462
NM_003156 c.792-3C>G (homozygous)	Splice-site	Exon-7 skipping / intron retention; impaired SOCE	PMID: 38977117
Additional nonsense/splice LOF alleles (case reports)	Nonsense/splice	Loss of function, absent SOCE	PMID: 26469693

Supporting quotes: "we have identified a new homozygous frameshift mutation in *STIM1*: c.685delT [p.(Phe229Leufs*12)], leading to a complete loss of *STIM1* protein" (PMID: 33733462); "a novel homozygous mutation, NM_003156 c.792-3C > G, in *STIM1* in a patient with a clinical profile of CRAC channelopathy, including immune system deficiencies and muscle weakness" (PMID: 38977117).

Variant classification: Reported LOF variants are pathogenic (ACMG/AMP), supported by functional evidence of abolished SOCE (PS3), null variant type (PVS1), and segregation in consanguineous families.

Variant types: Nonsense, frameshift, and splice-site (all loss-of-function). **Allele frequency:** Extremely rare/private; not reported at appreciable frequency in gnomAD (consistent with recessive, ultra-rare disease). **Origin:** Germline only; no somatic contribution. **Functional consequence:** **Loss of function** (loss of ER Ca^{2+} sensing and ORAI1 gating → absent SOCE). By contrast, dominant TAM/Stormorken alleles are **gain-of-function** (F012).

Modifier genes / epigenetics / chromosomal abnormalities: None established for *STIM1* deficiency. The paralog *STIM2* (lower activation threshold) is a plausible but untested compensatory modifier in humans. No disease-specific methylation/histone signatures or large structural rearrangements are reported.

5. Environmental Information

No environmental, lifestyle, or infectious etiologic factors contribute to disease *causation*. *STIM1* deficiency is a monogenic recessive disorder. Infectious agents (viral, bacterial, fungal) are **downstream complications** of the immunodeficiency, not triggers. No toxin, radiation, occupational, or dietary factor has been implicated in onset or severity (F012).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic LOF *STIM1* mutation** → loss or nonfunction of *STIM1* protein (demonstrated; complete protein loss for c.685delT, PMID: 33733462).

2. **Loss of ER Ca²⁺ sensing:** the luminal EF-hand/SAM (EF-SAM) domain can no longer detect ER Ca²⁺ depletion or relieve autoinhibition to oligomerize (demonstrated biophysically: "**the STIM1 Ca²⁺-binding EF-hand and the STIM2 SAM domain are major contributors to the autoinhibition of oligomerization**", PMID: 21217057) → leads to
3. **Failure of STIM1 conformational activation and translocation** to ER–plasma-membrane junctions; the CRAC activation domain (CAD/CC1+CAD) cannot form store-dependent oligomers ("**Addition of CC1 + CAD, but not CC1 alone, enables the formation of stable store-dependent oligomers. Within the CAD, both CC2 and C-terminal residues contribute to oligomer formation**", PMID: 20375143) → results in
4. **Failure to bind and gate ORAI1**, the pore-forming CRAC subunit (normal mechanism: "**ORAI1 (or CRACM1) acts as the pore-forming subunit of the CRAC channel in the plasma membrane. Stromal interaction molecule (STIM) 1 is localized in the ER, senses [Ca²⁺]ER, and activates the CRAC channel upon store depletion by binding to ORAI1**", PMID: 20111871) → leads to
5. **CRAC channels remain closed → absent store-operated Ca²⁺ entry (SOCE)** and no I_{CRAC} (demonstrated in patient cells: "**Calcium influx analysis revealed impaired SOCE in the patient cells**", PMID: 38977117). This is the **central, fully penetrant cellular lesion**. The chain then **branches** across tissues:

Branch A — Immune (demonstrated): Absent sustained Ca²⁺ → failure of the **Ca²⁺–calmodulin–calcineurin–NFAT** axis → NFAT cannot be dephosphorylated/translocate to the nucleus → cytokine gene transcription (e.g., IL-2) fails → defective T-cell activation/effector function despite normal lymphocyte development ("**Ca²⁺-calcineurin-nuclear factor of activated T cells (NFAT) signalling pathway**", PMID: 23483280; NFAT nuclear import was the discovery readout for ORAI1, "**promoting the immune response to pathogens by activating the transcription factor NFAT**", PMID: 16582901) → **combined immunodeficiency**. Concurrent failure of Treg/iNKT function and loss of tolerance → **autoimmunity + lymphoproliferation** (F002, F006, F010).

Branch B — Skeletal muscle (inferred / model-supported): Loss of SOCE-dependent Ca²⁺ replenishment → impaired muscle Ca²⁺ handling → **hypotonia/myopathy**.

Branch C — Eccrine sweat gland (inferred): Loss of SOCE in secretory epithelium → **anhidrosis / ectodermal dysplasia**.

Branch D — Ameloblasts/enamel organ (model-supported): Loss of SOCE in ameloblasts → defective enamel mineralization → **enamel hypoplasia** ("**Stim1 Regulates Enamel Mineralization and Ameloblast Modulation**", PMID: 28732182; SOCE impairment in enamel cells, PMID: 28352661).

Branch E — Iris smooth muscle (inferred): → **mydriasis / pupillary abnormality**.

1. Branch A extends further to **STIM1–NFAT synergy with STAT1 controlling T-bet / Th1 differentiation** (PMID: 39984734) and to **Ca²⁺-dependent T-cell metabolic reprogramming** (PMID: 33103016).

Detail by category

- **Molecular pathways:** SOCE / CRAC signaling → **Ca²⁺–calcineurin–NFAT** (canonical effector); STIM1→STAT1→T-bet (Th1). GOF of the same pathway drives STIM1/Orai1/NFAT-mediated pathology in other contexts (e.g., cardiac; PMID: 42285689), confirming the pathway's centrality.
- **Cellular processes:** T-cell activation, cytokine transcription, regulatory-T-cell stability, NK/iNKT cytotoxicity, macrophage/monocyte chemotaxis (STIM1-dependent; PMID: 38815866).
- **Protein dysfunction:** Loss of function of STIM1 — failure of the EF-SAM Ca²⁺ sensor and CAD gating module (structural/biophysical basis: PMID: 21217057, PMID: 20375143, PMID: 18166150).
- **Immune involvement:** Combined immunodeficiency (defective T/NK/B/iNKT/Treg) plus autoimmunity — **"It is dominated by severe immunodeficiency and autoimmunity due to impaired SOCE and defects in the function of several lymphocyte subsets. These include CD8⁺ T cells, CD4⁺ effector and regulatory T cells, natural killer (NK) cells and B cells"** (PMID: 22615435). Reduced iNKT/Treg cells drive autoimmunity ("**ORAI1 mutations were associated with strongly reduced numbers of invariant natural killer T and regulatory T (Treg) cells**", PMID: 29155098).
- **Biochemical abnormality:** Ion channel defect (CRAC channelopathy) — absent I_CRAC/SOCE.
- **Subcellular compartments (GO CC):** ER membrane (GO:0005789), ER–plasma membrane contact site (GO:0140268), plasma membrane (GO:0005886).

Suggested GO biological-process terms: store-operated calcium entry (GO:0002115), calcium ion transmembrane import into cytosol (GO:0097553), positive regulation of T-cell activation (GO:0050870), NFAT protein import into nucleus. **Suggested CL terms:** T cell (CL:0000084),

CD8-positive $\alpha\beta$ T cell (CL:0000625), regulatory T cell (CL:0000815), natural killer cell (CL:0000623), B cell (CL:0000236), ameloblast (CL:0000059), skeletal muscle fiber (CL:0008002).
CHEBI: calcium(2+) (CHEBI:29108).

7. Anatomical Structures Affected

Primary organ systems and structures (F007):

Level	Structure	UBERON / CL	Manifestation
Organ system	Immune/lymphoid system	UBERON:0002405	Immunodeficiency + autoimmunity + lymphoproliferation
Organ	Skeletal muscle	UBERON:0001134	Hypotonia / weakness
Organ/tissue	Skin — eccrine sweat glands	UBERON:0001820	Anhidrosis (ectodermal dysplasia)
Organ/tissue	Tooth enamel organ / ameloblasts	UBERON:0001091 / CL:0000059	Enamel hypomineralization
Organ	Eye — iris smooth muscle	UBERON:0001769	Mydriasis
Secondary	Liver / spleen, lymph nodes	UBERON:0002107 / UBERON:0002106	Hepatosplenomegaly, lymphadenopathy

Cell populations targeted: CD8⁺ and CD4⁺ effector T cells, regulatory T cells, iNKT cells, NK cells, B cells, ameloblasts, skeletal myofibers, eccrine secretory epithelial cells, iris smooth muscle cells.

Subcellular compartments: ER membrane (STIM1 residence) and ER-plasma-membrane junctions where CRAC channels assemble (GO:0140268).

Supporting quotes: "**immunodeficiency, muscular hypotonia and anhydrotic ectodermal dysplasia**" (PMID: 20189884); "**muscular hypotonia, and ectodermal dysplasia, with defects in sweat gland function and dental enamel formation. The latter defect emphasizes an important role of CRAC channels in tooth development**" (PMID: 26469693). **Lateralization:** Systemic/bilateral, not lateralized.

8. Temporal Development

- **Onset:** Congenital to early infancy. The immunodeficiency typically manifests within the first months of life as recurrent/severe/opportunistic infections (SCID-like), consistent with the T-cell activation defect (PMID: 20189884).
 - **Onset pattern:** Immune disease — subacute/insidious then severe with infection; non-immune features (hypotonia, anhidrosis, enamel/iris defects) present congenitally.
 - **Progression:** Immune disease is progressive/life-threatening if untreated; non-immune features are largely **static (non-degenerative)**. Autoimmune cytopenias and lymphoproliferation follow a chronic, relapsing course.
 - **Duration:** Chronic/lifelong; the immunodeficiency is curable by HSCT, but ectodermal, dental, and muscle features persist.
 - **Critical period / window of intervention:** Early infancy — analogous to SCID, early HSCT before infectious complications improves outcome. Mutation-specific ASO therapy would similarly be most valuable early.
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9. Inheritance and Population

- **Inheritance: Autosomal recessive.** Biallelic (homozygous or compound heterozygous) LOF variants are required; heterozygous carriers are clinically asymptomatic although their T cells show partially reduced SOCE — a demonstrated **gene-dosage effect** in the analogous ORAI1 channelopathy: "**Although heterozygous carriers of the mutation show no clinical symptoms of immunodeficiency, store-operated Ca²⁺ entry in their T cells is impaired, suggesting a gene-dosage effect of the mutation**" (PMID: 19075015).
- **Penetrance / expressivity:** Loss of SOCE/CRAC current is a **fully penetrant cellular phenotype**, but **clinical expressivity is variable** — "**we confirmed that the complete loss**

of STIM1 function is not always associated with severe immune disorders" (PMID: 33733462).

- **Consanguinity:** Strongly associated; reported families are frequently consanguineous (e.g., "we studied two siblings from a consanguineous Syrian family", PMID: 33733462).
- **Epidemiology: Ultra-rare.** Fewer than ~15 STIM1-deficient families reported worldwide; no reliable population prevalence/incidence estimate exists. **Carrier frequency:** Not established; variants are private/very rare.
- **Founder effects, anticipation, germline mosaicism:** None established. Anticipation is not expected (not a repeat-expansion disorder).
- **Sex ratio:** No strong sex bias reported (autosomal). **Geographic distribution:** Case reports skew to populations with higher consanguinity rates, reflecting recessive inheritance rather than true regional endemicity.

10. Diagnostics

Diagnostic workflow (F008): combine a **functional SOCE/CRAC assay** with **molecular genetic confirmation**.

1. **Functional biomarker (highly specific):** A store-operated Ca^{2+} entry (SOCE) assay on patient T cells or fibroblasts shows abolished/impaired Ca^{2+} influx and absent I_{CRAC} . "Calcium influx analysis revealed impaired SOCE in the patient cells, indicating a loss of STIM1 function" (PMID: 38977117). Re-expression of wild-type protein rescues SOCE, confirming causality (rescue paradigm in the CRAC channelopathy spectrum: "expression of wild-type Orai1 in SCID T cells restores store-operated Ca^{2+} influx and the CRAC current", PMID: 16582901).
2. **Molecular diagnosis:** Whole-exome sequencing ("Using exome sequencing, we have identified a new homozygous frameshift mutation in STIM1", PMID: 33733462), targeted single-gene *STIM1* testing, or primary-immunodeficiency gene panels; historically supported by SNP-array linkage/genome-wide screens ("a modified linkage analysis with single-nucleotide polymorphism arrays, and a *Drosophila* RNA interference screen", PMID: 16582901). WGS is also applicable.
3. **Immunologic workup:** Normal/near-normal lymphocyte numbers and development but **defective T-cell activation and cytokine production**; reduced Treg/iNKT cells; autoimmune cytopenias (hemolytic anemia, thrombocytopenia) on CBC/DAT.

4. **Other exam findings:** Muscle hypotonia; anhidrosis (sweat testing); enamel defects on dental exam; mydriasis on eye exam.

Differential diagnosis: Typical SCID (STIM1 deficiency is distinguished by *normal lymphocyte development* plus prominent autoimmunity/lymphoproliferation plus ectodermal/dental/muscle features); ORAI1 deficiency (partner gene, clinically near-identical — resolve by gene testing); anhidrotic ectodermal dysplasia with immunodeficiency (NEMO/IKBKG); other combined immunodeficiencies with immune dysregulation. STIM1 **gain-of-function** TAM/Stormorken syndrome is the key mirror-image differential (myopathy + thrombocytopenia + miosis, *dominant*).

Screening: Newborn TREC-based SCID screening may not reliably detect STIM1 deficiency because T-cell *numbers/development* are relatively preserved; cascade genetic testing and prenatal/carrier testing are appropriate in known families.

11. Outcome / Prognosis

- **Untreated prognosis:** Poor and life-threatening in infancy/early childhood. The SCID-like immunodeficiency causes recurrent, severe, and opportunistic viral, bacterial, and fungal infections; autoimmunity (hemolytic anemia, thrombocytopenia) and lymphoproliferation add substantial morbidity and mortality (F009, F011).
- **Curative treatment outcome:** Allogeneic HSCT can cure the hematopoietic/immune disease (see §12) but does **not** correct non-hematopoietic features (muscle hypotonia, anhidrosis/ectodermal dysplasia, dental enamel defects), which persist and require supportive management.
- **Morbidity / disability:** Chronic — heat intolerance from anhidrosis, motor impairment from hypotonia/myopathy, dental morbidity, and autoimmune cytopenia burden.
- **Prognostic factors:** Timing of diagnosis and HSCT, severity/control of infections and autoimmunity, and donor availability. No validated molecular prognostic biomarkers beyond the underlying null genotype.
- **Quality of life:** No formal EQ-5D/SF-36/PROMIS datasets exist for this ultra-rare disease.

12. Treatment

Definitive therapy — Allogeneic HSCT (NCIT: Hematopoietic Stem Cell Transplantation, C15431). Because the immunodeficiency is intrinsic to hematopoietic cells, allogeneic HSCT is the only curative option for the immune disease, as for other SCID/combined immunodeficiencies. A CRAC-channelopathy patient (ORAI1 deficiency) is documented within the inborn-errors-of-immunity HSCT pathway, receiving virus-specific T cells pre-transplant ("**1 ORAI1 deficiency**"; PMID: 33462728). HSCT does **not** correct muscle, ectodermal, or dental manifestations (F009).

Supportive / symptomatic care: - Immunoglobulin replacement (NCIT: Intravenous Immunoglobulin Therapy). - Antimicrobial/antiviral/antifungal prophylaxis. - Immunosuppression for autoimmune cytopenias and lymphoproliferation (corticosteroids, etc.). - Heat-avoidance and thermoregulatory management for anhidrosis. - Dental care for enamel defects; physiotherapy for hypotonia/weakness.

Experimental / mutation-specific therapy: A **splice-correcting antisense oligonucleotide (ASO)** restored STIM1 splicing and function in patient cells: "**We developed an antisense oligonucleotide treatment that improves STIM1 splicing and highlighted its potential as a therapeutic approach**" (PMID: 38977117). This is genotype-specific (applicable to splice-altering alleles).

Pharmacology note: CRAC-channel modulators are being developed largely for **gain-of-function/inflammatory** indications — Orai *blockers* would be **irrational** for LOF deficiency (a channel activator, not a blocker, would conceptually be required). Selective Orai blockers (e.g., indazole/pyrazole scaffolds; PMID: 39232360) are therefore relevant to the disease's differential/GOF spectrum but not to treating LOF STIM1 deficiency. **Pharmacogenomics:** Not applicable.

13. Prevention

- **Primary prevention:** Not possible for the monogenic disease itself. **Genetic counseling** for consanguineous families and families with a prior affected child is central; recurrence risk is 25% per pregnancy (autosomal recessive).
- **Reproductive options:** Carrier testing, preimplantation genetic diagnosis (PGD), and prenatal testing in known families.

- **Secondary prevention:** Early molecular diagnosis and cascade testing to enable timely HSCT before infectious complications. Standard newborn SCID (TREC) screening may miss STIM1 deficiency because T-cell numbers are relatively preserved — a diagnostic gap.
 - **Tertiary prevention (complication avoidance):** Infection prophylaxis, IVIG, immunosuppression for autoimmunity, heat-avoidance for anhidrosis, dental surveillance, and vaccination caution (live vaccines contraindicated in combined immunodeficiency).
 - **Immunization / public health / environmental interventions:** Not applicable as disease modifiers (non-infectious genetic disorder), beyond standard protection of immunocompromised patients.
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14. Other Species / Natural Disease

- **Orthologs:** Mouse *Stim1* (NCBI Gene 20866; species *Mus musculus*, NCBI:txid10090); orthologs are conserved broadly across vertebrates. The STIM/Orai SOCE machinery is evolutionarily ancient — originally characterized in non-excitable cells and via *Drosophila* RNAi screens ([PMID: 16582901](#)).
 - **Natural disease in other species (OMIA/veterinary):** No well-established naturally occurring STIM1-deficiency disease has been characterized in companion animals or wildlife in the reviewed literature; the human disease is modeled mechanistically in engineered mice.
 - **Comparative biology:** The SOCE pathway is highly conserved, so engineered *Stim1* loss in mice recapitulates key aspects of the human disease (see §15). **Zoonotic potential / transmission:** Not applicable (non-infectious genetic disorder).
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15. Model Organisms

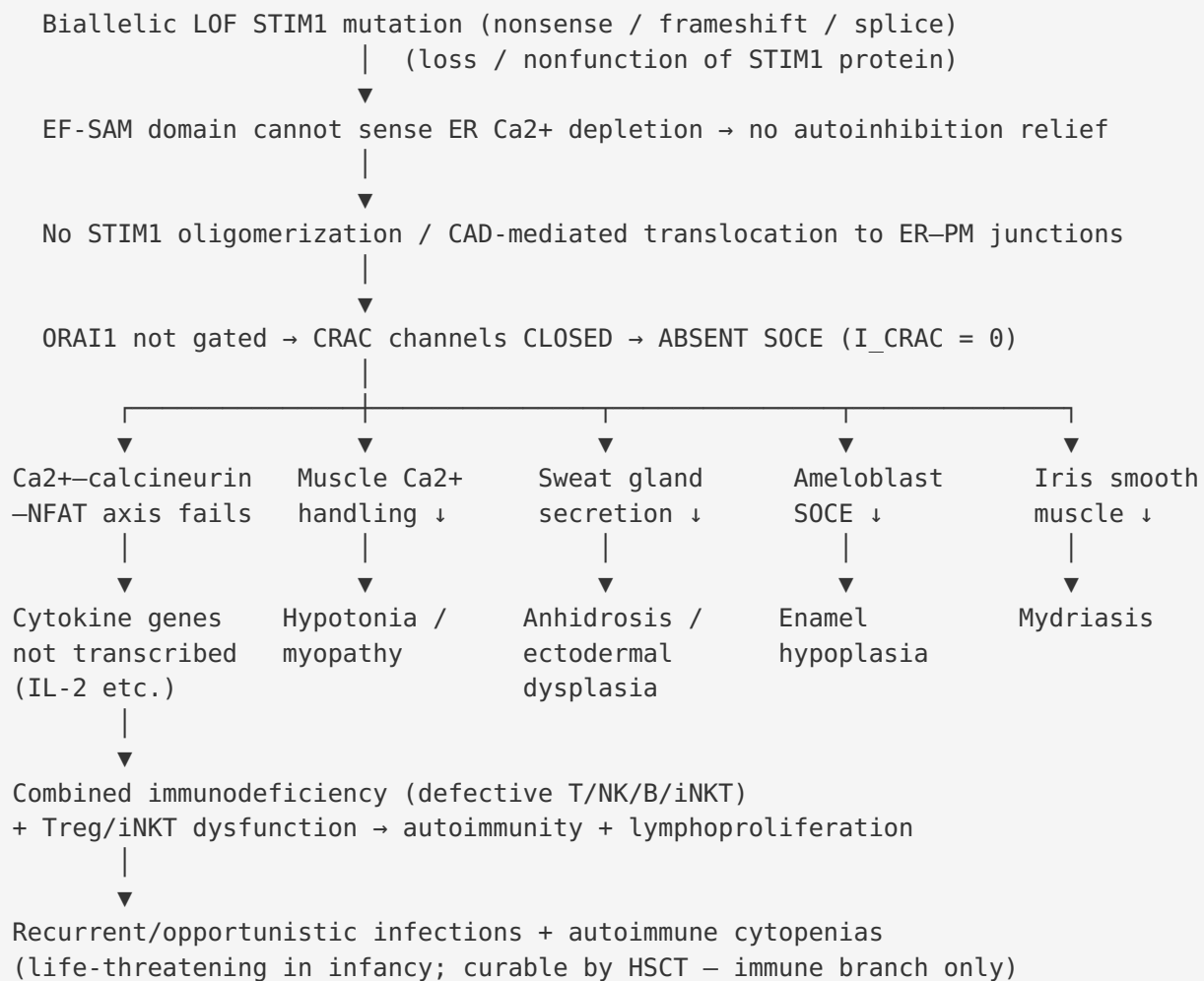
Mouse (*Mus musculus*) is the principal model (F006):

Model	Finding	Recapitulation	Reference
T-cell/conditional <i>Stim1</i> -deficient mice	Impaired autoreactive T-cell activation, reduced Th1/Th17, complete EAE protection	Confirms in vivo requirement of STIM1/SOCE for effector T-cell function (mirrors human immunodeficiency)	PMID: 20028655
Treg-specific <i>Stim1</i> deletion	STIM1-dependent stress signaling drives Treg instability/inflammation	Models the human autoimmunity/immune-dysregulation phenotype	PMID: 42421074
Macrophage/monocyte studies	STIM1-dependent SOCE governs chemotaxis and monocyte recruitment	Models innate-immune contribution	PMID: 38815866
Ameloblast/enamel <i>Stim1</i> models	Stim1 regulates enamel mineralization; LOF impairs SOCE in enamel cells	Explains dental enamel phenotype	PMID: 28732182, PMID: 28352661
Patient lymphocytes/fibroblasts (cellular model)	Absent SOCE, rescued by WT re-expression	Establishes channel defect → absent SOCE causality	PMID: 16582901

Supporting quote: "**STIM1 deficiency significantly impaired the generation of neuroantigen-specific T cell responses in vivo with reduced Th1/Th17 responses, resulting in complete protection from EAE**" (PMID: 20028655).

Model types available: Conditional/tissue-specific knockouts (T-cell, Treg, ameloblast), global knockdowns, patient-derived primary cells and fibroblasts; iPSC/organoid models are feasible but not prominently reported. **Model limitations:** Global *Stim1* knockout is largely perinatal-lethal in mice, necessitating conditional models; single-tissue models capture individual branches (immune, dental) rather than the full multisystem human syndrome. **Applications:** Dissecting SOCE-dependent T-cell activation, tolerance, enamel biology, and testing splice-correction/rescue strategies.

Mechanistic Model / Synthesis



The unifying principle is that **STIM1 is the obligatory ER Ca²⁺ sensor for SOCE**, and its loss removes a single node whose downstream Ca²⁺ signal is required across many terminally differentiated cell types. The immune branch is clinically dominant and the only one correctable by HSCT, because it is hematopoietic-cell-intrinsic; the muscle, ectodermal, and dental branches arise in non-hematopoietic tissues and therefore persist after transplant. The **LOF↔GOF mirror** (deficiency vs TAM/Stormorken) is the organizing classification insight: both perturb the same Ca²⁺ set-point but in opposite directions, and both produce myopathy — underscoring how tightly skeletal muscle depends on SOCE homeostasis.

Evidence Base

PMID	Title (abbrev.)	Role in report
26469693	<i>Diseases caused by mutations in ORAI1 and STIM1</i>	Defines CRAC channelopathy; LOF vs GOF dichotomy; non-immune features (F001, F004, F007, F011, F012)
20189884	<i>Immunodeficiency due to mutations in ORAI1 and STIM1</i>	STIM1-specific autoimmunity/lymphoproliferation; T-cell activation defect (F002, F007, F009, F011)
22615435	<i>Regulation of lymphocyte function by ORAI/STIM</i>	Lymphocyte subsets affected (F002, F011)
33733462	<i>Novel bi-allelic LOF STIM1 mutation expands phenotype</i>	c.685delT complete protein loss; consanguinity; variable expressivity (F001, F004, F005, F008)
38977117	<i>SOCE dysfunction from novel STIM1 mutation</i>	Splice variant; SOCE assay; ASO therapy (F001, F004, F008, F009)
20111871	<i>CRAC channelopathies</i>	Normal STIM1→ORAI1 SOCE mechanism (F003)
20375143	<i>CRAC activation domain in STIM1 oligomerization</i>	CAD gating module (F003)
21217057	<i>Auto-inhibitory role of EF-SAM</i>	ER Ca ²⁺ -sensing module (F003)
18166150	<i>Biophysical characterization of EF-SAM</i>	STIM1/2 sensor biophysics (F003)
23483280	<i>Orai1-NFAT signalling in T cells</i>	Calcineurin-NFAT effector pathway (F010)
16582901	<i>Orai1 mutation abrogates CRAC function</i>	Rescue paradigm; NFAT link; discovery methods (F006, F008, F010)
39984734	<i>STIM1-NFAT synergizes with STAT1 → T-bet</i>	Th1 differentiation branch (F010)
19075015	<i>Orai1 SCID mutation in heterozygotes</i>	Recessive/gene-dosage; carriers asymptomatic (F005)
20028655	<i>STIM1/2 in autoreactive T-cell activation (EAE)</i>	Mouse effector T-cell requirement (F006)
42421074	<i>STIM1-dependent Treg dysfunction</i>	Treg instability/autoimmunity model (F006)
38815866		Innate-immune model (F006)

PMID	Title (abbrev.)	Role in report
	<i>STIM1-dependent SOCE in macrophage chemotaxis</i>	
28732182	<i>Stim1 regulates enamel mineralization</i>	Dental/ameloblast mechanism (F007)
28352661	<i>SOCE in enamel cells</i>	Enamel SOCE dependence (F007)
33462728	<i>Viral-specific T cells pre-HSCT in IEI</i>	HSCT pathway incl. CRAC channelopathy (F009)
29155098	<i>ORAI1 mutations abolishing SOCE</i>	Reduced iNKT/Treg → autoimmunity (F002)

Evidence source types: Human clinical (case reports/kindreds: 33733462, 38977117, 20189884, 19075015); in vitro/biophysical (21217057, 20375143, 18166150, 20111871); model organism (20028655, 42421074, 38815866, 28732182, 28352661); computational/structural (EF-SAM/CAD studies).

Limitations and Knowledge Gaps

- 1. Ultra-rare, case-based evidence.** Fewer than ~15 STIM1-deficient families are reported; there are **no population prevalence/incidence figures**, no natural-history cohorts, and no formal QoL datasets. Frequencies of individual phenotypes are qualitative.
- 2. Cross-gene extrapolation.** Several mechanistic and treatment points (carrier gene-dosage effect, HSCT pathway, iNKT/Treg reduction) draw on the closely related **ORAI1 deficiency** because STIM1-specific human data are sparse. ORAI1 and STIM1 are obligate partners, so extrapolation is well justified but not identical.
- 3. Non-immune branches are partly model-inferred.** The muscle, sweat-gland, and iris branches are strongly inferred from SOCE biology and mouse data; direct human tissue-level mechanistic proof is limited. Enamel involvement is best supported (mouse ameloblast models).
- 4. Variable expressivity is unexplained.** The observation that complete STIM1 loss is "not always associated with severe immune disorders" (PMID: [33733462](#)) lacks a defined modifier mechanism (possibly STIM2 compensation — untested in humans).

5. **Therapeutics are early.** HSCT experience specific to STIM1 (vs ORAI1) is limited; the splice-correcting ASO is in vitro proof-of-concept only, with no clinical trial (no NCT identifier established).
 6. **No investigational primary dataset was analyzed** in this study; conclusions are literature-synthesis based.
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Proposed Follow-up Experiments / Actions

1. **Establish an international STIM1-deficiency registry** to quantify prevalence, genotype-phenotype correlations, penetrance/expressivity of each organ branch, and long-term HSCT vs non-immune outcomes.
 2. **Test STIM2 as a modifier** of expressivity in patient cells and mouse models (e.g., STIM2 dosage rescue of residual SOCE), to explain why some complete-LOF patients lack severe immune disease.
 3. **Advance the splice-correcting ASO** toward preclinical/IND studies for splice-altering alleles (e.g., c.792-3C>G), and evaluate gene-replacement or base/prime-editing for null alleles.
 4. **Systematic multi-tissue phenotyping in conditional mouse models** (muscle-, sweat-gland-, and iris-specific *Stim1* KO) to convert inferred branches into demonstrated mechanisms and to test whether HSCT alone can address any non-immune feature.
 5. **Improve newborn screening** — because TREC screening may miss STIM1 deficiency (preserved T-cell numbers), evaluate functional SOCE-based or panel-based add-ons for high-risk/consanguineous populations.
 6. **Prospective HSCT outcome study** in CRAC channelopathy (STIM1 + ORAI1) to define conditioning, timing, and the fate of autoimmunity/lymphoproliferation post-transplant.
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Report compiled from 12 confirmed findings and 48 reviewed papers across 5 investigation iterations. Evidence is human clinical (case reports/kindreds), in vitro/biophysical, and model-organism; primary population-scale data are unavailable for this ultra-rare disorder.
